



INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Pizza Store Analysis

By

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About me :

Damini Mukund Borude

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Why I Want to Learn Data Science:

I am eager to learn Data Science because I am passionate about exploring data-driven technologies and understanding how data can be used to make informed decisions and solve real-world problems.

Work Experience:

I have my internship at Innomatics research labs in Data Science and GenAI, in which I have contributed to some projects that helped me to develop my analytical and technical skills.

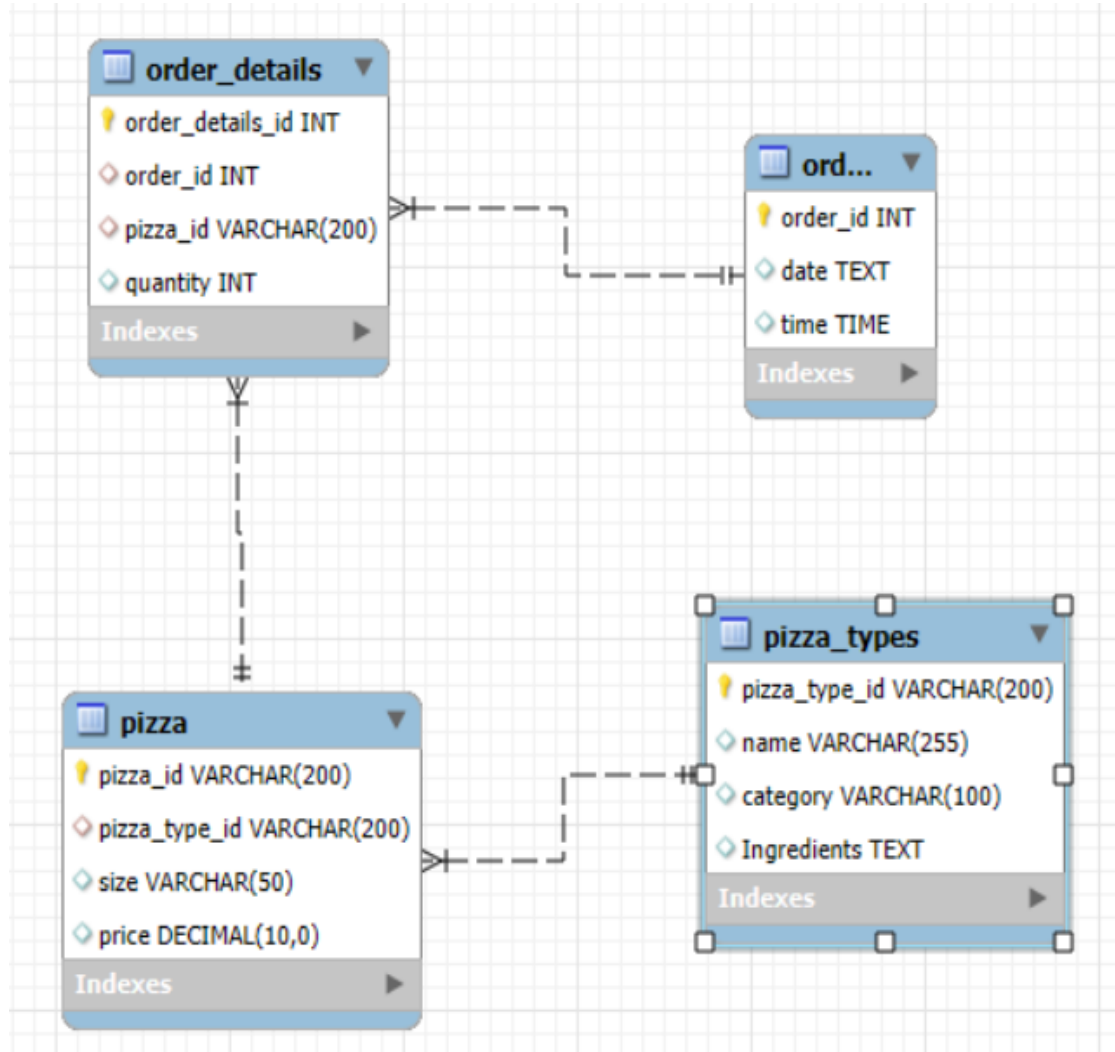
Profiles:

- LinkedIn: <https://www.linkedin.com/in/daminiborude/>
- GitHub: <https://github.com/daminiborude>

Objective of the Project :

- Analyze overall sales performance to understand total orders, revenue, and customer buying patterns.
- Identify top-performing pizzas and categories based on sales quantity and revenue contribution.
- Evaluate customer preferences by examining the most popular pizza sizes, categories, and ingredients.
- Determine peak business hours and daily trends to help in resource planning and staff scheduling.
- Calculate revenue distribution and growth over time to assess financial performance.
- Provide actionable business insights and recommendations to improve sales, optimize inventory, and enhance marketing strategies.

EER Diagram and Schema :



- **Orders table:** It gives information like order ID, date, and time. Each order can include multiple pizzas, which are recorded in the Order_Details table.
- **Order_Details table:** It acts as a link between Orders and Pizzas, it stores which pizzas were ordered and in what quantity.
- **Pizza table:** It contains details such as pizza ID, size, and price. Each pizza also belongs to a specific type, which is defined in the Pizza_Types table.
- **Pizza_Types table:** It stores information like the pizza name, category, and ingredients.

Key Areas of Exploration :

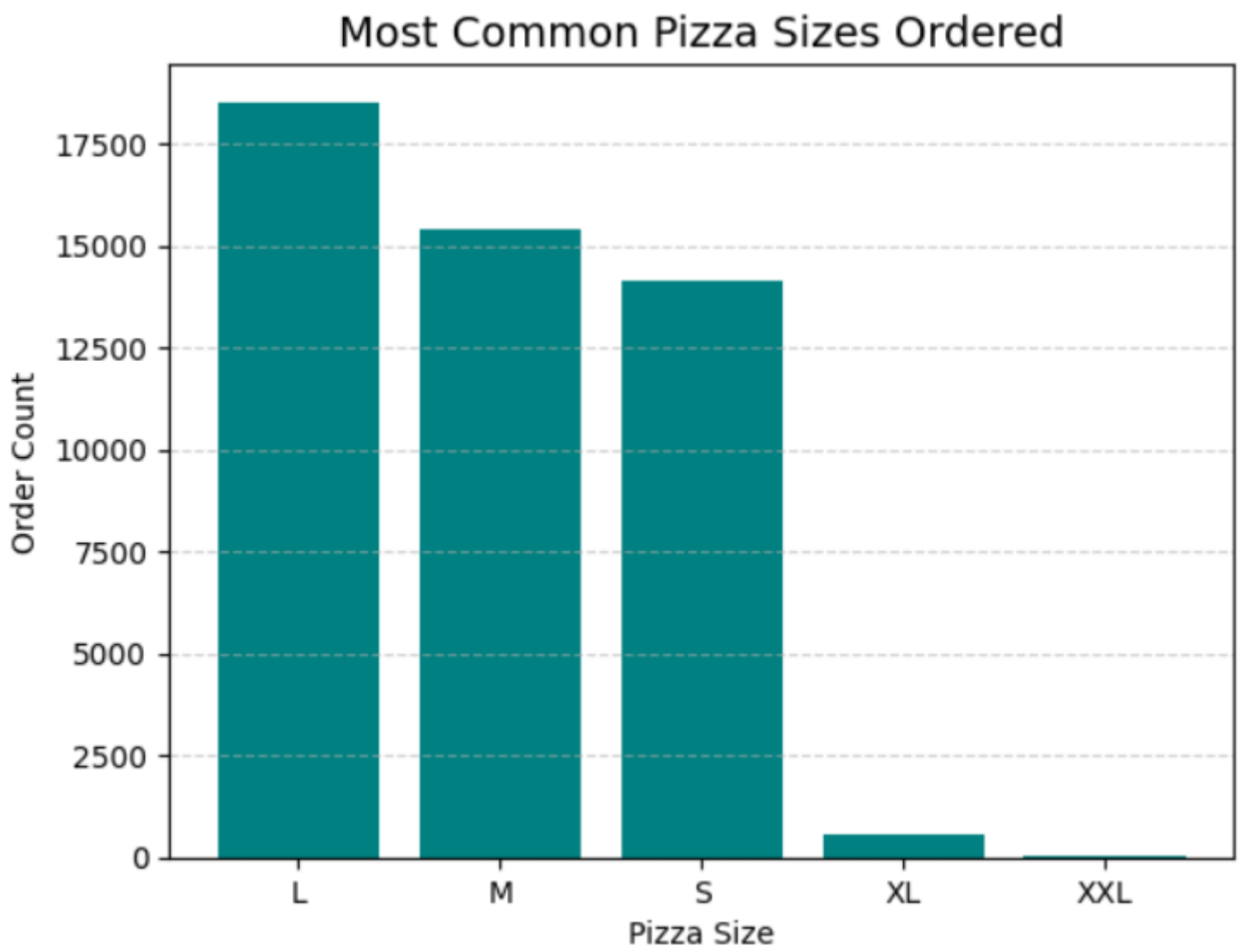
- Identify the most common pizza size ordered.
- List the top 5 most ordered pizza types along with their quantities.
- Find the total quantity of each pizza category ordered.
- Determine the distribution of orders by hour of the day.
- Find the category-wise distribution of pizzas (count of pizza types per category).

Identify the most common pizza size ordered.

```
select p.size, count(*) as order_count
from order_details od
join pizza p on od.pizza_id = p.pizza_id
group by p.size
order by order_count desc;
```

Result Grid

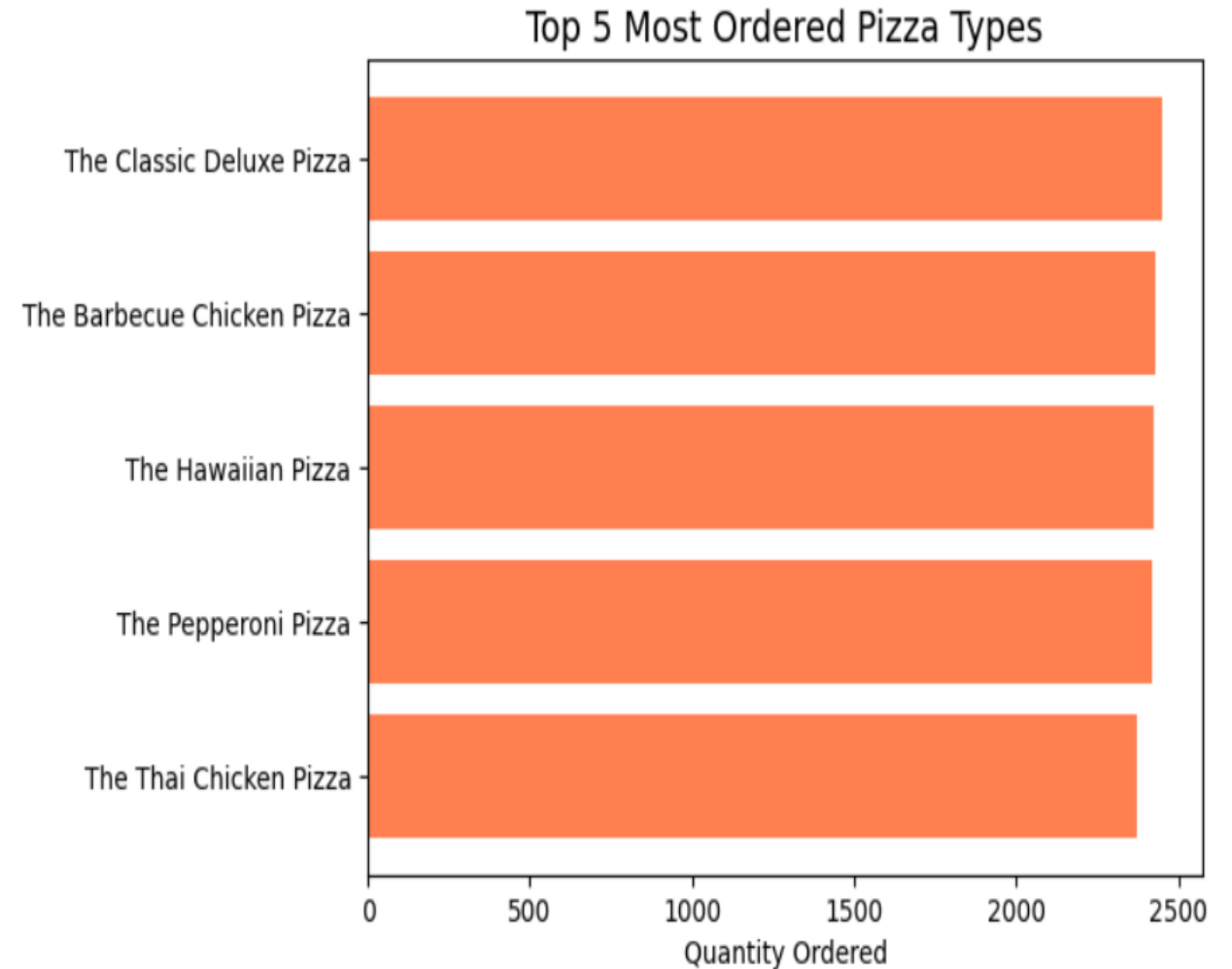
	size	order_count
▶	L	18526
	M	15385
	S	14137
	XL	544
	XXL	28



List the top 5 most ordered pizza types along with their quantities.

```
select pt.name, SUM(od.quantity) AS total_quantity
from order_details od
join pizza p on od.pizza_id = p.pizza_id
join pizza_types pt on p.pizza_type_id = pt.pizza_type_id
group by pt.name
order by total_quantity desc
limit 5;
```

	name	total_quantity
▶	The Classic Deluxe Pizza	2453
	The Barbecue Chicken Pizza	2432
	The Hawaiian Pizza	2422
	The Pepperoni Pizza	2418
	The Thai Chicken Pizza	2371

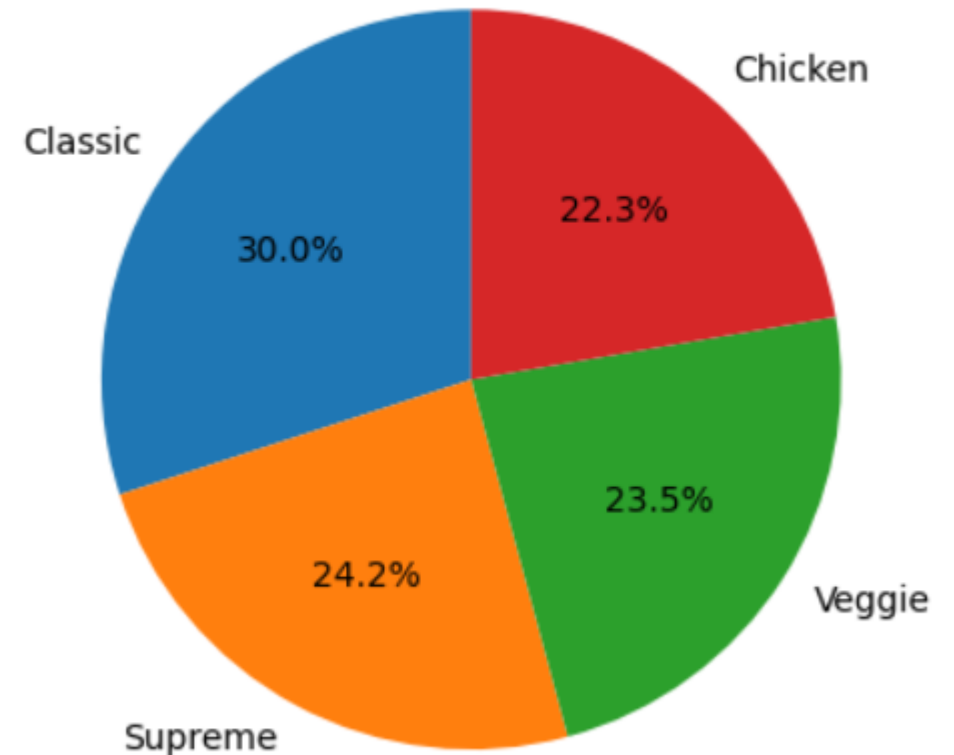


Find the total quantity of each pizza category ordered.

```
select pt.category, SUM(od.quantity) AS total_quantity
from order_details od
join pizza p on od.pizza_id = p.pizza_id
join pizza_types pt on p.pizza_type_id = pt.pizza_type_id
group by pt.category
order by total_quantity desc;
```

	category	total_quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

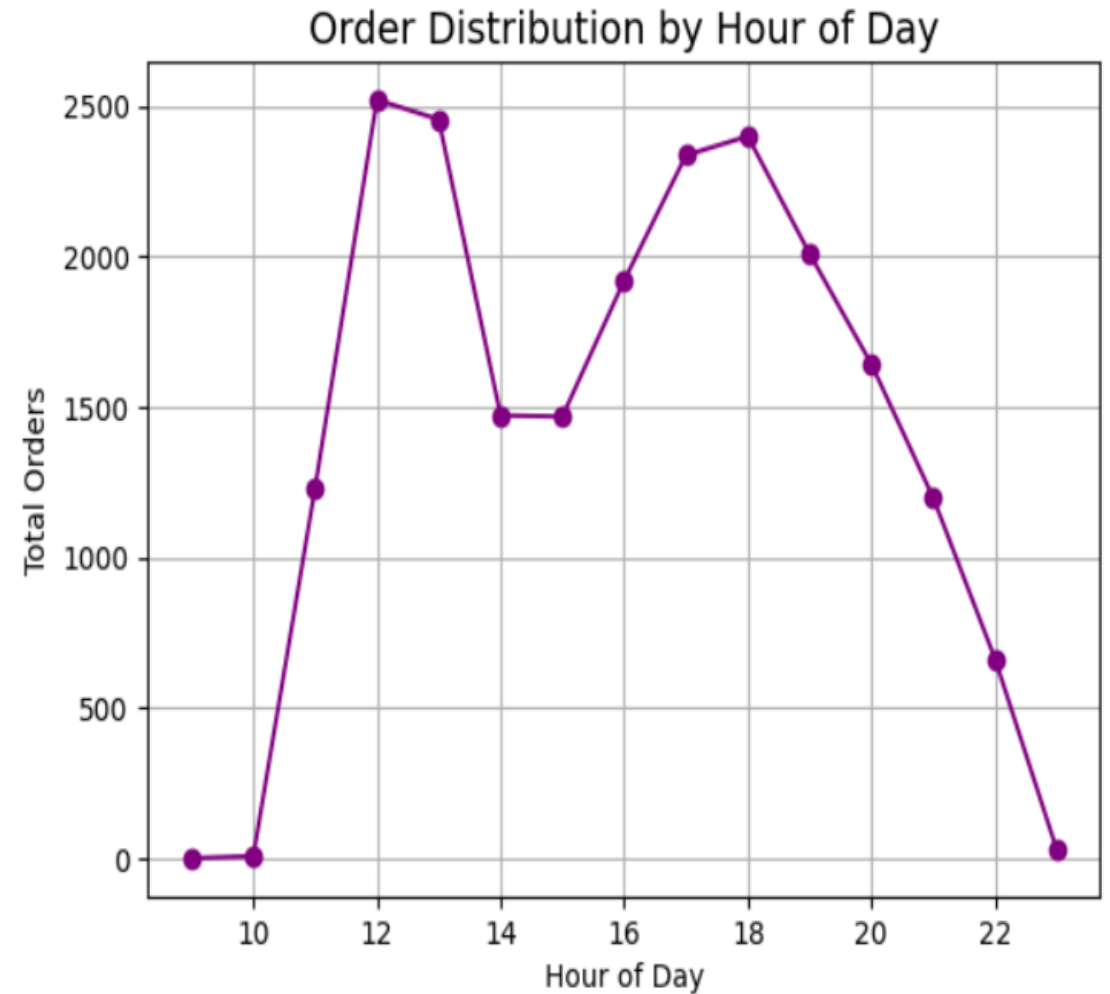
Pizza Category Distribution (%)



Determine the distribution of orders by hour of the day.

```
select hour(time) AS order_hour, COUNT(order_id) as total_orders
from orders
group by order_hour
order by order_hour;
```

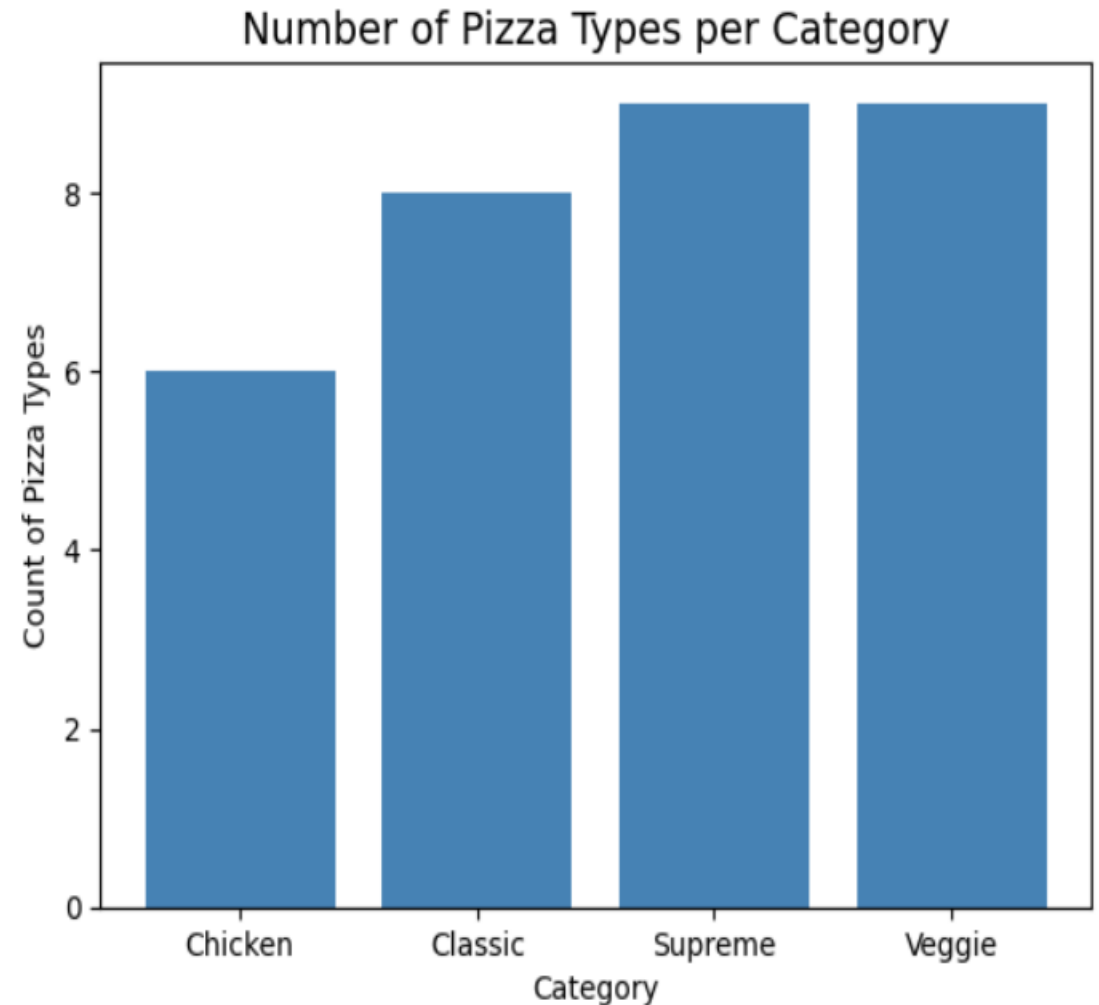
	order_hour	total_orders
▶	9	1
	10	8
	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
	18	2399
	19	2009
	20	1642
	21	1198
	22	663
	23	28



Find the category-wise distribution of pizzas (count of pizza types per category).

```
select category, count(name) as count_pizza_type
from pizza_types
group by category
order by count_pizza_type;
```

	category	count_pizza_type
►	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9



Conclusion :

- **Identified Key Customer Preferences**

Medium and Large pizzas were most preferred, especially from the Classic category showing people love traditional flavors and shareable sizes.

- **Discovered Peak Business Hours**

Most orders came between 6 PM and 9 PM, helping the shop owner plan staff, inventory, and offers during high-demand times.

- **Supported Better Business Decisions**

The analysis helped the owner understand sales trends, manage stock efficiently, and focus on top-performing pizzas.

- **Turned Data into Actionable Insights**

The project proved how data analysis can turn simple sales records into powerful insights that improve profit and customer satisfaction.

Experiences and Challenges :

Experiences:

- Gained practical knowledge of writing complex SQL queries to extract, transform, and analyze large pizza sales datasets.
- Learned to design and create multiple related tables with proper primary and foreign key constraints ensuring data integrity.

Challenges:

- Faced issues while importing CSV files into MySQL due to path errors, Large amount of data and formatting mismatches, requiring data cleaning and correct delimiter handling.
- Some aggregate and join queries initially took longer to execute, optimized them using indexing.
- Encountered missing values and duplicate entries in order details, which required careful validation and correction for accurate results.



THANK
YOU

